

Aneesh Maganti

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EDUCATION

New York University, Tandon School of Engineering, Brooklyn, NY Fall 2024
Bachelor of Science, Computer Science GPA: **3.80**
Relevant Courses: Algorithmic Machine Learning, Offensive Security, ML Visualization, Computer Architecture

SKILLS

Languages	C++, Python, Javascript, C#, Java, Bash
Technologies	PyTorch, Next.js, SDL, QT, React, Node, PostgreSQL, Sklearn, Docker, Linux

EXPERIENCE

Affirm, New York City, NY, *Software Engineer, Traffic Engineering* July 2025 - Present

- Orchestrated the migration of External Secrets Operator (ESO) from v0.7.2 to v0.20.2 across 72 Kubernetes clusters, leveraging a Controller Class mechanism to run multiple concurrent operators and manage diverse secret types including Affirm service, infra, and internal secrets.
- Rewrote a core Kubernetes build tool in Rust, slashing Helm chart build times by 78% (from 14 minutes to 3 minutes); successfully drove team-wide adoption of the tool and mentored external contractor to execute over 40 high-velocity deployments.
- Spearheaded the Cert-Manager upgrade and modernised secret management by developing Terraform configurations for dynamic key generation and implementing agent-driven update flows

Qualcomm, San Diego, CA, *Software Engineer Intern, AI Software/ML Group* May 2024 - Aug 2024

- Implemented various memory optimizations across the Qualcomm AI Runtime (QAIRT), reducing peak memory usage by 20% across benchmarked ResNet50 and LLAMA2 and LLAMA3 LLM models
- Optimized AIMET quantization pipeline via C++ PyBind API to reference static graph tensors within Python subprocess, reducing in-memory allocations by 50%
- Refactored QAIRT/AIMET toolchain via PyTorch 'meta' device to prevent unnecessary weight initialization and modified model export to use in-memory IR graph, reducing disk space and improving runtime

New York University, Brooklyn, NY, *Teaching Assistant (Machine Learning)* Sep 2023 - Dec 2023

- Instructed students weekly for machine learning topics of written and programming tasks through office hours
- Graded weekly assignments on the basis of proper algorithm implementation, code correctness and style

NYU Algorithms and Foundations Group, Brooklyn, NY, *ML Academic Researcher* Feb 2023 - Sep 2023

- Designed and implemented *Deltagonalshift*, a diagonal estimator for a dynamic matrix under Professor Christopher Musco to improve neural network optimization based on DeltaShift trace estimation algorithm
- Demonstrated Deltagonalshift was more effective than repeatedly running Hutchinson's diagonal estimator

PROJECTS/ACTIVITIES

BUGS Open Source Club President Sep 2022 - Dec 2023

- Started and led 50+ member club by coordinating biweekly workshop and project coding events to discuss software engineering skills, foster contributions to open source, and create a fun, inclusive CS community.
- Led multiple workshops including discussion of open-source licenses, JavaScript Playwright automation, and an overview of emulation and the internals of C++ Game Boy emulator.
- Developed Next.js-React website NYU Syllabi with Netlify hosting and Docusaurus-based NYU CS Wiki websites

Dot Matrix - Game Boy Emulator August 2023

- Designed x86 C++ emulator for the Game Boy platform by implementing 255 standard + 240 cb instructions
- Simulated hardware features including registers, graphics (SDL), memory, timers, interrupts, and input handling